

TATHAGATA ROY CHOWDHURY

Assistant Professor • PhD Research Scholar

Artificial Intelligence • Healthcare Informatics • Machine Learning • Explainable AI

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- 🌐 [Portfolio](#)
- 🆔 orcid.org/0000-0003-1052-4628
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- 📖 [Web of Science Researcher Profile](#)
- 🎓 [Semantic Scholar Profile](#)

Research Interests

Artificial Intelligence • Healthcare Informatics • Machine Learning • Deep Learning • Federated Learning • Explainable AI • Computer Vision • Medical Image Analysis • Precision Oncology

Professional Profile

An Assistant Professor with more than ten years of experience in Computer Science and Engineering, currently pursuing a PhD at the National Institute of Technology Silchar. My research focuses on Artificial Intelligence for Healthcare, Explainable AI, Medical Image Analysis, Federated Learning, Deep Learning and Clinical Decision Support Systems. I have authored books, published research in international venues, including Springer, contributed to edited volumes, served as a reviewer for multiple IEEE conferences and Elsevier journals, and actively mentored undergraduate research projects. My objective is to develop trustworthy, secure, and explainable AI systems capable of addressing real-world healthcare challenges while fostering excellence in teaching, research, and academic leadership.

Research Highlights

10+ Years of Teaching Experience	PhD Research Scholar (NIT Silchar)
Assistant Professor	AI in Healthcare Researcher
Springer Author	Multiple Books Authored
IEEE Conference Reviewer	Elsevier Reviewer
Indian Patent Applicant	UK Registered Design Contributor
Oracle Academy Authorized Trainer	Academic Coordinator and Department In-Charge
Healthcare AI	Explainable AI & Federated Learning

Core Research Interests

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Healthcare Informatics
- Medical Image Analysis
- Explainable Artificial Intelligence (XAI)
- Federated Learning
- Computer Vision
- Precision Oncology
- Clinical Decision Support Systems
- Bioinformatics
- Data Security and Privacy

Personal Information

Full Name	Tathagata Roy Chowdhury
Current Position	Assistant Professor, Techno Engineering College Banipur
Research Status	PhD Research Scholar, NIT Silchar
Nationality	Indian
Date of Birth	29 March 1990
Email	tathagatamtech13@gmail.com
Website	https://www.tutoringcs.com
Mobile	+91-8902503176
Research Areas	AI, Healthcare Informatics, Machine Learning, XAI

Career Objective

To contribute to internationally recognised research in Artificial Intelligence, Healthcare Informatics, Medical Image Analysis, Explainable Artificial Intelligence, Federated Learning and Clinical Decision Support Systems through high-quality research, innovative teaching, interdisciplinary collaboration and academic leadership. My long-term goal is to develop trustworthy AI solutions that improve patient care while mentoring future researchers and contributing to global scientific advancements.

Academic Appointments

Assistant Professor – Techno Engineering College Banipur

Duration: March 2024 – Present

Teaching undergraduate Computer Science courses, supervising student research, curriculum development, mentoring projects, organising technical events and conducting research in the fields of AI and Healthcare Informatics.

Assistant Professor In-Charge – St. Mary's Technical Campus

Duration: September 2022 – March 2024

Managed departmental academic activities, faculty coordination, student mentoring, accreditation support, workshop organisation and departmental administration.

Assistant Professor – Brainware University

Duration: July 2022 – September 2022

Delivered lectures in Computer Science, programming and Machine Learning while developed academic content and laboratory exercises.

Assistant Professor – Elite College of Engineering

Duration: September 2019 – June 2022

Taught C, Java, Python, Data Structures, Software Engineering and Artificial Intelligence-related subjects. Guided projects and laboratory sessions were conducted.

Assistant Professor & Academic Advisor – Chaibasa Engineering College

Duration: February 2015 – September 2019

Academic advising, curriculum planning, technical event organisation, workshop coordination, and student mentoring.

Administrative Responsibilities

- Department In-Charge
- Academic Coordinator
- Student Mentor
- Workshop and FDP Coordinator
- ERP Development and Management (ORKA ERP)
- Academic Content Developer
- Research Supervisor for Undergraduate Projects
- Oracle Academy Authorized Trainer

Professional Strengths

- More than ten years of teaching experience.
- Extensive experience in Artificial Intelligence and Healthcare research.
- Published author with Springer and international publishers.
- Reviewer for multiple IEEE international conferences.
- Active researcher in Explainable AI and Federated Learning.
- Strong academic leadership and curriculum development experience.
- Expertise in Python, Deep Learning, Computer Vision and Medical AI.

Educational Qualifications

Degree	Institution / University	Year	Result
PhD (Pursuing)	National Institute of Technology Silchar	2024–Present	Ongoing
M.Tech (CSE)	University of Calcutta	2014	65.42%
B.Tech (CSE)	Saroj Mohan Institute of Technology (WBUT)	2011	CGPA 7.49
Higher Secondary (Science)	WBCHSE	2007	67%
Madhyamik	WBBSE	2005	66%

Research Areas

Major Research Domains

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- Artificial Intelligence
- Machine Learning
- Deep Learning
- Explainable AI (XAI)
- Federated Learning
- Healthcare Informatics
- Medical Image Analysis
- Computer Vision
- Precision Oncology
- Clinical Decision Support
- Bioinformatics
- Secure AI Systems

Teaching Portfolio

Core Subjects Taught

- Programming in C
- Object Oriented Programming using Java
- Python Programming
- Data Structures
- Design and Analysis of Algorithms
- Machine Learning
- Artificial Intelligence
- Data Mining
- Compiler Design
- Digital Image Processing

Research Expertise

Research Competencies

- Development of AI models for healthcare diagnosis.
- Deep learning architectures for medical image analysis.
- Explainable Artificial Intelligence for trustworthy prediction.
- Federated Learning for privacy-preserving healthcare.
- Cancer diagnosis using Machine Learning.
- Data preprocessing, feature engineering and synthetic oversampling.
- Research methodology, academic writing and peer review.

Technical Skills

Programming	Python, Java, C, C++, PHP
AI Frameworks	TensorFlow, PyTorch, Keras, OpenCV
Databases	MySQL, Oracle, SQL Server
Research Tools	LaTeX, SPSS, WEKA, RapidMiner, Jupyter Notebook
Web Technologies	HTML, CSS, JavaScript, PHP
Operating Systems	Windows, Linux
Version Control	Git, GitHub

Academic Philosophy

My academic philosophy is centred on integrating high-quality teaching with impactful research. I believe in outcome-based education, interdisciplinary collaboration, Research-led teaching and continuous innovation. My goal is to inspire students to become ethical engineers, independent researchers and lifelong learners while developing Artificial Intelligence solutions that address real-world healthcare challenges.

Selected Research Publications

Recent Publications

1. Advanced DeepLungCareNet: A Next-Generation Framework for Lung Cancer Prediction. Springer LNNS (2026).
2. Comparative Analysis of Feature Extraction Techniques for Breast Cancer Classification using Support Vector Machines.
3. Precision Colon Cancer Screening using Machine Learning Techniques.
4. Cardiac Arrest Prediction using Machine Learning.
5. Rice Disease Detection using Deep Learning Techniques.
6. Deep Learning Models: A Comprehensive Review.

Books Authored

- A Pathway to J.E.C.A. (2023)
- Anomaly Detection in Log Files (2024)

- Unveiling the Black Box: Deep Learning and Explainable AI (2024)
- A Complete Overview on Web Technology (2026)

Book Chapters

- AI-Driven Environmental Sustainability (Wiley + IEEE Press)
- AI-powered OCR and Healthcare Informatics in Industry 5.0
- Multiple international book chapters in Artificial Intelligence, Healthcare Informatics and Machine Learning.

Patents and Intellectual Property

Indian Patent

Title: A Privacy-Preserving and Secured Federated Learning Framework for Decentralized Genomic Data Analysis
 Application No.: 202631052210
 Status: Filed (2026)

UK Registered Design

Blockchain-Secured Health Data Retrieval Device
 Registered Design No. 6528955
 United Kingdom Intellectual Property Office

International Reviewing Activities

- IEEE EDIGE 2K26 – Technical Reviewer
- IEEE ICESAIA 2026 – Technical Reviewer
- IEEE ICMACC 2026 – Technical Reviewer
- IEEE INCODE 2026 – Technical Reviewer
- IEEE TEMSMET 2026 – Technical Reviewer
- Elsevier Journal Reviewer
- Analytics Global Conference Reviewer

Research Projects

- DeepLungCareNet
- AFWE-CGFS Framework
- Strategic Synthetic Oversampling (SSO)
- Bio-Cipher Federated Learning Framework
- AI-based PCOS Prediction System
- Breast Cancer Diagnostic Framework

Professional Memberships

- IEEE
- ACM
- IAENG
- Computer Society of India (CSI)
- Oracle Academy Authorized Trainer

Awards and Recognition

- Best Online Teacher Award
- Oracle Academy Authorized Trainer
- 5-Star Educator (UrbanPro)
- First Prize – Tabla Competition

Academic Profiles

Google Scholar

ORCID

ResearchGate

Semantic Scholar

LinkedIn

Web of Science

Declaration

I hereby declare that the information furnished in this curriculum vitae is true and correct to the best of my knowledge and belief.

Tathagata Roy Chowdhury
Assistant Professor
PhD Research Scholar